

Hw 1

① The two experiments that I chose are

①.. Water Jet falling on a metal cylinder
The idea is to observe how water jet breaks into drops.

When the water jet falls on the end of the cylinder, it can be seen how the water jet breaks up into drops.

The results that I saw were really similar to the ones on the video, it were recreated with a metal rod and, to experiment a little bit more, I also used other surfaces such a metal spoon and a plastic rod. I need to say that the results are better observed when using a rod and that can be a little difficult to observe.

②.. Dyed mixture of alcohol and water.

The main idea of this experiment is to observe the surface tension pumping action in detail.

When we swirl the mixture, we can observe a thin film on the side of the glass, the evaporation of alcohol causes it to have a lower proportion of alcohol, so there is a higher surface tension on the walls compared to the liquid in glass.

This makes the liquid move up to the high surface tension region. Since it accumulates on the top, it starts to drop to the liquid again. The experiment that I did was the same, with same results.

② My favorite dimensionless number is

Reynolds number.

It is the number that I have worked the most in classes like transport phenomena.

It is interesting that Reynolds number measures the importance of inertia vs viscosity in a flowing fluid.

If the number is smaller, the viscous forces dominates.

If the number is high, the inertial forces dominates.

I think it is a really cool number because it is a dimensionless ratio and helps us to predict flow regime.

③ Characteristic lengths with surface tension or the definition.

- The capillary length $l_{\Gamma\psi} = \left(\frac{\Gamma}{\psi}\right)^{\frac{1}{2}}$

- Capillary rise height $\Gamma \sim g \rho h D$

- Thickness of a liquid film $l_{E\Gamma} = \left(\frac{E}{\Gamma}\right)^{\frac{1}{2}}$ $E = \text{Hamaker constant}$

2). Butterfly Proboscis.

→ Capillary rise

To calculate the capillary rise we use Jurin's Height equation

$$H = \frac{2\gamma \cos \theta_E}{\rho g R} \rightarrow R = \frac{2\gamma \cos \theta_E}{\rho g H}$$

The assumptions made to calculate the radius

- Complete wetting of the tube meaning that $\theta = 0^\circ \therefore \cos \theta_E = 1$
- Liquid (water) at room temperature
where $\gamma = 0.72 \text{ N/m}$ or 72 mN/m
 $\rho = 1000 \text{ kg/m}^3$

$$\text{since } N = \frac{\text{kg} \cdot \text{m}}{\text{s}^2}$$

$$R = \frac{2(0.72 \frac{\text{N}}{\text{m}})}{1000 \frac{\text{kg}}{\text{m}^3} \cdot 9.81 \frac{\text{m}}{\text{s}^2} \cdot H} \rightarrow R = \frac{0.144 \frac{\text{N}}{\text{m}}}{9800 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}} \cdot H}$$

$\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \cdot \frac{\text{m}}{\text{kg}} = \frac{\text{m}^2}{\text{s}^2}$

$$\rightarrow H = 1 \text{ mm or } 0.001 \text{ m}$$

$$R = \frac{0.144 \frac{\text{N}}{\text{m}}}{9800 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}} \cdot (0.001 \text{ m})} \rightarrow R = 0.0147 \text{ m or } 1.47 \text{ cm}$$

$$\rightarrow H = 10 \text{ mm or } 0.01 \text{ m}$$

$$R = \frac{0.144 \frac{\text{N}}{\text{m}}}{9800 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}} (0.01 \text{ m})} \rightarrow R = 0.00147 \text{ m or } 1.47 \text{ mm}$$

$$\rightarrow H = 100 \text{ mm or } 0.1 \text{ m}$$

$$R = \frac{0.144 \frac{\text{N}}{\text{m}}}{9800 \frac{\text{kg}}{\text{m}^2 \cdot \text{s}} (0.1 \text{ m})} \rightarrow R = 0.000147 \text{ m or } 0.147 \text{ mm}$$

To calculate the Hydrostatic at different Heights

$$P = \rho g h$$

At $H = 0.1 \text{ mm}$ or 0.0001 m

$$P = \left(1000 \frac{\text{kg}}{\text{m}^3}\right) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) (0.0001 \text{ m}) \rightarrow P = 0.981 \text{ Pa}$$

At $H = 5 \text{ mm}$ or 0.005 m

$$P = \left(1000 \frac{\text{kg}}{\text{m}^3}\right) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) (0.005 \text{ m}) \rightarrow P = 49.05 \text{ Pa}$$

At $H = 50 \text{ mm}$ or 0.05 m

$$P = \left(1000 \frac{\text{kg}}{\text{m}^3}\right) \left(9.81 \frac{\text{m}}{\text{s}^2}\right) (0.05 \text{ m}) \rightarrow P = 490.5 \text{ Pa}$$

Compared to the Atmospheric pressure $P_a = 101,325 \text{ Pa}$

H	Hydrostatic pressure	% of Atmospheric pressure
0.1 mm	0.981 Pa	$9.68 \times 10^{-4} \%$
5 mm	49.05 Pa	0.048%
50 mm	490.5 Pa	0.48%

Even at 50 mm deep the hydrostatic pressure is less than 0.5 % of the atmospheric pressure.

3).

The vertical force from surface tension along a contact line is

$$F = \gamma \pi R^2 H$$

where $H = \frac{2\gamma \cos \theta_E}{\rho g R}$

substituting

$$F = \gamma \pi R^2 \left(\frac{2\gamma \cos \theta_E}{\rho g R} \right)$$

$$F = 2\pi R \cdot \gamma \cdot \cos \theta_E$$

For the insect leg that is hydrophobic, the contact angle is large, so it can be considered $\sin \theta_E = 1$

The weight of the liquid column is

$$W = 2\pi R \gamma$$

• if the surface tension is large, it can support a larger weight.

→ To obtain the upper limit to the weight that can be supported by a cylindrical rod placed on water we could use the equation from before replacing πR by L which is the length of the rod.

$$W = 2 \cdot \gamma \cdot L$$

b) -

$$F_{\text{Euler}} = \left(\frac{\pi}{2}\right)^2 \frac{ER^4}{L^2}$$

Buckling occurs when the compressive force exceeds the Euler's critical load

$$F_{\text{Euler}} = F \rightarrow F = 2\pi R \cdot \gamma$$

$$2\pi R \cdot \gamma = \left(\frac{\pi}{2}\right)^2 \frac{ER^4}{L^2}$$

Solving for $L_c = L$

$$L^2 = \frac{\pi^2}{4} \frac{ER^4}{2\pi R \cdot \gamma}$$

$$L = \sqrt{\frac{\pi ER^3}{8\gamma}}$$

Compared to the capillary length $L_c = \sqrt{\frac{\gamma}{\rho g}}$

It can be seen that "Buckling length depends on elasticity"
"Capillary length depends on gravity"

→ To estimate the critical length for human hair

$E = 4 \text{ GPa}$, $R = 40 \text{ micron} \rightarrow 4 \times 10^{-5} \text{ m}$
water $\gamma = 0.072 \frac{\text{N}}{\text{m}}$

$$L_c = \sqrt{\frac{\pi (4 \times 10^9 \text{ Pa}) (4 \times 10^{-5} \text{ m})^3}{8 (0.072 \frac{\text{N}}{\text{m}})}}$$

$$\frac{\text{Pa} \cdot \text{m}^3 \cdot \text{m}}{\text{N}} \quad \frac{\text{Pa} \cdot \text{m}^4}{\text{N}} \rightarrow \frac{\cancel{\text{N}}}{\cancel{\text{m}^2}} \cdot \frac{\cancel{\text{m}}^4}{\cancel{\text{N}}} = \text{m}^2$$

$$1 \text{ Pa} = 1 \frac{\text{N}}{\text{m}^2}$$

$$L_c = 0.037 \text{ m or } 3.7 \text{ cm}$$

The cheerios effect

The equation for the interface $z = h(x)$

Assumptions

- Small interfacial slope
- Incompressible liquid

$$z = h(x)$$

from the young-Laplace $\Delta p = \gamma \kappa$

for a small slope $\kappa \approx \frac{d^2 h}{dx^2}$

And $\Delta p = \text{hydrostatic pressure}$ that is equal to $\Delta p = \rho g h$

$$\rho g h = \gamma \cdot \frac{d^2 h}{dx^2} \quad \leftarrow \text{equation 1}$$

$$\text{since } L_c \equiv \sqrt{\gamma / \rho g} \rightarrow L_c^2 = \gamma / \rho g$$

$$\eta = \frac{\gamma}{\rho g} \cdot \frac{d^2 h}{dx^2} \rightarrow h = L_c^2 \cdot \frac{d^2 h}{dx^2} \rightarrow \frac{h}{L_c^2} = \frac{d^2 h}{dx^2}$$

$$\text{from } \frac{d^2 h}{dx^2} = \frac{h}{L_c^2}$$

Linear ODE

$$h'' - \frac{h}{L_c^2} = 0$$

The characteristic equation is

$$r^2 - \frac{1}{L_c^2} = 0 \rightarrow r = \pm \frac{1}{L_c}$$

General equation

$$h(x) = A e^{x/L_c} + B e^{-x/L_c}$$

On region 1 ($x < 0$)

$$h(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow -\infty$$

$$\text{since } x \rightarrow -\infty, e^{-x/L_c} \rightarrow \infty$$

$$\text{so } h_1(x) = A_1 e^{x/L_c}$$

Because of the geometry

$$h'_1(0) = \cot \theta_1$$

$$h'_1(x) = \frac{A_1}{L_c} e^{x/L_c}$$

$$\text{At } x=0 \rightarrow \frac{A_1}{L_c} = \cot \theta_1$$

$$A_1 = L_c \cot \theta_1$$

$$h_1(x) = A_1 e^{x/L_c} \rightarrow h_1(x) = L_c \cot \theta_1 e^{x/L_c}$$

On region 3 ($x > d$)

$$h(x) \rightarrow 0 \quad \text{as} \quad x \rightarrow \infty$$

$$h_3(x) = B_3 e^{-x/L_c}$$

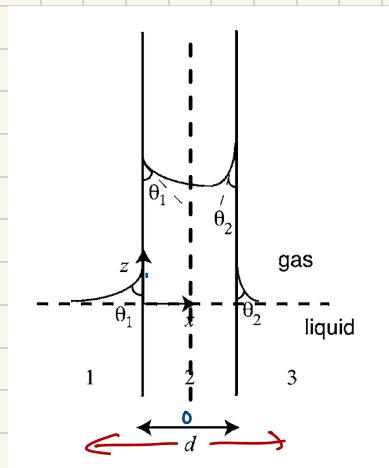
$$h'_3(d) = -\cot \theta_2$$

$$h'_3(x) = \frac{-B_3}{L_c} e^{-x/L_c}$$

$$\text{At } x=d \rightarrow \frac{-B_3}{L_c} e^{-d/L_c} = -\cot \theta_2$$

$$B_3 = L_c \cot \theta_2 e^{d/L_c}$$

$$h_3(x) = L_c \cot \theta_2 e^{d-x/L_c}$$



In region 2 (middle)

$$h_2(x) = A_2 e^{x/L_c} + B_2 e^{-x/L_c}$$

$$\underline{x=0} \rightarrow h_2'(0) = -\cot \theta_1$$

$$h_2'(x) = \frac{A_2}{L_c} e^{x/L_c} - \frac{B_2}{L_c} e^{-x/L_c}$$

$$\frac{A_2 - B_2}{L_c} = -\cot \theta_1 \rightarrow A_2 - B_2 = -L_c \cot \theta_1$$

$$h_2'(d) = \cot \theta_2 \rightarrow A_2 - B_2 = L_c \cot \theta_2$$

$$A_2 e^{d/L_c} - B_2 e^{-d/L_c} = L_c \cot \theta_2$$

$$h_2(x) = \frac{L_c}{\sinh(d/L_c)} \left[\cot \theta_1 \cosh\left(\frac{d-x}{L_c}\right) + \cot \theta_2 \cosh\left(\frac{x}{L_c}\right) \right]$$

→ Characteristic separation distance

$$\text{in solution 1, we obtain } L_c = \sqrt{\frac{\gamma}{\rho g}}$$

From expression

$$h = e^{-x/L_c}$$

→ $F_h(d) = e^{-d/L_c}$ meaning that interaction force decays exponentially with separation.

According to the article

L_c for water is $L_c = 2.7 \text{ mm}$ so the attraction will disappear after a few mm.

→ Dimensionless group

In expression 6 we can observe the dimensionless parameter

$$\frac{d}{L_c}$$

But i think that the relevant dimensionless scale is capillary length

→ Changes of temperature

⑤ Size of a puddle formed by a large drop on a solid substrate
Capillary length is exceeded

$$R \gg \kappa^{-1}$$

Since large drops where radius R exceeds κ^{-1} , the gravitational effect dominates.

At eq. it takes the shape of a liquid pancake with thickness e .

There are two forces involved and can be used to calculate e by expressing the equilibrium of the horizontal forces acting on a portion of the liquid.

① Surface forces, $\gamma_{so} - (\gamma + \gamma_{sl})$

② Hydrostatic pressure $\bar{p}$ integrating over the entire thickness of the liquid

$$\bar{p} = \int_0^e \rho g (e - \bar{z}) d\bar{z} = \frac{1}{2} \rho g e^2$$

Equilibrium of forces per unit length:

$$\frac{1}{2} \rho g e^2 + \gamma_{so} - (\gamma + \gamma_{sl}) = 0 \quad \text{leading to} \quad S = -\frac{1}{2} \rho g e^2$$

Young's Law describes the equilibrium of forces acting on the line of contact

$$\gamma_{so} - (\gamma \cos \theta_E + \gamma_{sl}) = 0$$

$$\kappa^{-1} = \sqrt{\frac{\gamma}{\rho g}}$$

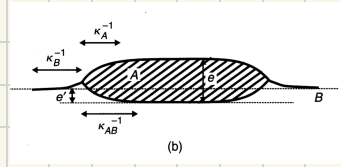
$$\therefore \gamma \cdot (1 - \cos \theta_E) = \frac{1}{2} \rho g e^2$$

$$2 \frac{\gamma}{\rho g} \underbrace{(1 - \cos \theta_E)}_{\sin \theta_E} = e^2 \quad \rightarrow \quad 2 \sqrt{\frac{\gamma}{\rho g}} \sin\left(\frac{\theta_E}{2}\right) = e$$

$$2 \kappa^{-1} \sin\left(\frac{\theta_E}{2}\right) = e$$

Size of puddle formed by a large drop of a floating lens formed on an immiscible liquid substrate.

For large lenses ($R \gg \kappa_{AB}^{-1}, \kappa_A^{-1}$)



Thickness, e , can be calculated by minimizing the energy, which can be rewritten in terms of thickness of the puddle. and the immersed depth e' ,

$$F(e) = \underbrace{\left[\frac{1}{2} \rho_A g (e - e')^2 \right]}_{\text{Energy of the drop above interface}} - \underbrace{\left[\frac{1}{2} (\rho_B - \rho_A) g e'^2 \right]}_{\text{Buoyancy}} - \underbrace{S}_{\text{surface energy}} \frac{\Omega}{e} \rightarrow \text{surface area}$$

using Archimedes condition

$$\rho_A g e = \rho_B g e' \quad \rightarrow \quad e' = \frac{\rho_A}{\rho_B} e$$

$$\text{Defining } \bar{\rho} = \frac{\rho_A}{\rho_B} \cdot (\rho_B - \rho_A)$$

$$F(e) = \left[\frac{1}{2} \bar{\rho} g e^2 - S \right] \frac{\Omega}{e} \quad \frac{1}{x} \quad x^{-1-1} = -\frac{1}{x^2}$$

Differentiating

$$F(e) = e \left[\frac{1}{2} \bar{\rho} g e - \frac{S}{e} \right] \quad \rightarrow \quad \frac{dF}{de} = e \left[\frac{1}{2} \bar{\rho} g + \frac{S}{e^2} \right]$$

$$0 = e \left[\frac{1}{2} \bar{\rho} g + \frac{S}{e^2} \right] \quad \rightarrow \quad 0 = \frac{1}{2} \bar{\rho} g + \frac{S}{e^2} \quad \rightarrow \quad -\frac{S}{e^2} = \frac{1}{2} \bar{\rho} g$$

$$-S = \frac{1}{2} \bar{\rho} g e^2$$

$$e = \sqrt{\frac{2 \cdot S}{\bar{\rho} g}}$$